Econometrics I

Lecture 7: Instrumental Variables II — weak instruments, LATE, judges, shift-share

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Where we are

- Session 6: why OLS fails under endogeneity, and how an instrument fixes it when everyone has the same β.
- Today, three things applied work cannot do without:
  1. Checking the assumptions, and what happens when the instrument is weak — including what changed in the advice since 2020.
  2. What IV estimates when effects differ across people: the local average treatment effect, and who the compliers are.
  3. The two designs behind most credible IV papers of the last fifteen years: judges and examiners, and shift-share exposure.
- Simulations with known answers throughout; Assignment 4 uses the last two designs.

Checking the assumptions; weak instruments

Checking The IV Assumptions I

- The two major assumptions:
  1. Relevance: $Cov(X, Z) \neq 0$, or that $\mathbb{E}\left[\mathbf{z}_i \mathbf{x}_i'\right]$ has full rank
  2. Exogeneity: All instruments must be uncorrelated with ε
- First, let's talk about the relevance assumption.
- It's easy to check that the sample analog to $Cov(X, Z) \neq 0$, or the analogous matrix condition.
- The more practically relevant concern is that the covariance will be *close* to zero. This is called **weak instruments**.
- How do we test and deal with weak instruments in practice?

- In finite samples there's a possibility that the sample covariance

$$\sum_{i=1}^{N}(x_i - \bar{x})(z_i - \bar{z})$$

  is close to zero, especially if the true covariance is relatively small.

- Dividing by zero is *bad*.

- Note we don't have the same problem in OLS as long as we have variation in the regressors.

Weak Instruments II

- There is **big** problem with IV estimation: 2SLS performs **terribly** in small samples
- Intuition: Instrumental variables relies on $Corr(X, Z) \neq 0$
  - If $Cov(X, Z) = 0$ then IV estimator is infinite!
- In practice, very rarely will $Cov(X, Z) = 0$ in data (just randomness)
  - Hence, IV will usually "work" with any choice of X and Z
  - Realistically, sometimes $Cov(X, Z)$ is small if Z only explains a small portion of X
- If $Cov(X, Z)$ is small we say that Z is a weak instrument

Weak Instruments and Small Sample Bias

Suppose that Z is *a bad instrument* in that $Cov(Z, \varepsilon)$ are correlated

- Similar to OVB

$$\hat{\beta}_1^{IV} \rightarrow \beta_1 + \frac{Cov(Z, \varepsilon)}{Cov(Z, X)} \times \frac{\sigma_\varepsilon}{\sigma_X}$$

- Compare to OLS case (biased because of OVB):

$$\hat{\beta}_1^{OLS} \rightarrow \beta_1 + Cov(X, \varepsilon) \times \frac{\sigma_\varepsilon}{\sigma_X}$$

- Which estimator is better here?
  - IV's advantage is $Corr(Z, \varepsilon)$ should be smaller than $Corr(X, \varepsilon)$
  - But if instrument is weak $Corr(Z, X) \approx 0$, then bias blows up even if $Corr(Z, \varepsilon)$ small
- Danger of weak instruments:
  - Very small bias in Z can be greatly magnified if instruments are weak
  - Amplified bias may have different sign than OLS

To do inference we rely on the CLT:

$$\hat{\beta}^{IV} \sim \mathcal{N}\left(\beta, Var\left(\hat{\beta}^{IV}\right)\right)$$

- First issue: the Normal Approximation fails unless N is huge! (the CLT does not work well)
  - Intuition: $\hat{\beta}^{IV} = Cov(Y, Z)/Cov(X, Z)$.
  - If $Cov(X, Z) \approx 0$ then small changes in estimated covariance can have huge effects on estimated coefficients

- Second issue: even if the normal approximation works the variance explodes!
  - Intuition from homoscedastic case: $Var(\hat{\beta}^{IV})) = \sigma^2/(N\sigma_X^2 R_{XZ}^2)$
  - If the R^2 of X on Z is very small, then variance is very large

What are we to do if instruments are weak?

- Easiest: get more instruments or better instruments
- Harder: adjust how to do inference
  - Turns out that if instruments are weak, asymptotic distribution of $\hat{\beta}$ will be the *ratio* of two *correlated* normal random variables
  - We will not pursue this (uses LIML)
- The good news: We can test $Cov(X, Z)$ by regressing $x_i = \pi_0 + \pi_1 Z_i + \eta_i$
  - Since Z and X are data, we can just see if $\hat{\pi}_1$ is far from zero
- **Rule of Thumb:** Only use instrument if t-stat of null hypothesis that $\pi_1 = 0$ is ≥ 10 (much bigger than stat significance of 1.96!)

$F > 10$ Cutoff

Some clarification:

- With only one endogenous variable, the $F > 10$ cutoff refers to the first-stage regression of that endogenous variable on all the exogenous variables. The null hypothesis for that F-statistic is that the coefficients on all the excluded instrumental variables are zero.
- With multiple endogenous variables, there are multiple first-stage regressions, and we should be wary if *any* of them suggests a weak instrument problem.
- Many have argued that the $F > 10$ cutoff is not stringent enough. See, for example, https://arxiv.org/abs/2010.05058Lee, McCrary, Moreira, and Porter

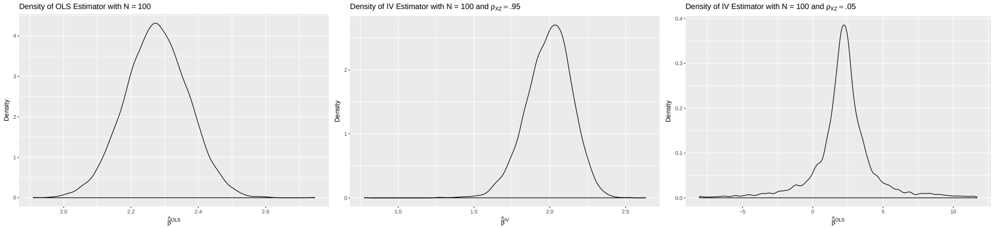
The model:

$$
\begin{aligned}
Y_i =& \beta_0 + \beta_1 X_i + \varepsilon_i \\
X_i =& \pi_0 + \pi_1 Z_i^{(1)} + \pi_2 Z_i^{(2)} + ... + \pi_m Z_i^{(m)} + \eta_i
\end{aligned}
$$

- $\pi \approx 0 \Rightarrow$ because of *sampling error*, there is a good chance that $\hat{\pi} < 0$ even if $\pi > 0$
- If $\hat{X}$ is the wrong *sign* or close to *zero* then it is very likely that $\hat{\beta}^{IV}$ will behave poorly
- **NB:** As $N \to \infty$, $\hat{\pi} \to \pi$, so weak instruments are a *small sample problem*.

$$Y = 1 + 2X + \varepsilon$$

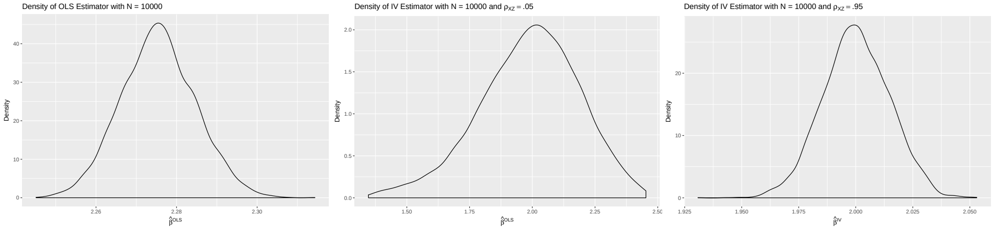
Figure 1: Distribution of $\widehat{\beta}$ for OLS, strong IV, weak IV with $N = 100$



- OLS is biased with $\mu = 2.27$ and $\sigma^2 = .008$
- Strong IV is unbiased (asymptotically) with $\mu = 1.99$ and $\sigma^2 = .024$
- Weak IV is unbiased (asymptotically) with $\mu = 2.05$ but $\boldsymbol{\sigma^2 = 1653.11}$

$$Y = 1 + 2X + \varepsilon$$

Figure 2: Distribution of $\widehat{\beta}$ for OLS, weak IV, strong IV with $N = 10000$



- OLS remains biased with $\mu = 2.28$ and $\sigma^2 = .000085$
- Strong IV is consistent with $\mu = 1.99$ and $\sigma^2 = .00021$
- Weak IV is consistent with $\mu = 2.05$ and even with $N = 10k$, $\sigma^2 = .05$

What is the Danger of the CLT Failing?

- With weak IV, CLT fails and behaves poorly even if N is large
- Remember that for testing we want to reject the null hypothesis incorrectly 5% of the time
  - When the CLT fails, our trusty 1.96 critical value is way off base
  - Example rejection probabilities for test that $\hat{\beta} = 2$

Hypothetical Situation	$P(\|t\| > 1.96)$
Weak IV, $N = 100$	0.3%
Strong IV, $N = 100$	4.2%
Weak IV, $N = 10000$	3.2%
Strong IV, $N = 10000$	5.1%

  - Goal should be 5% but for weak IV with N small, almost never reject; still only 3% even for N large—this is a bad test!

Example in R: Checking Angrist & Evans Validity

```r
 # First Stage Test in Angrist & Evans
flm1 = feols(morekids ~ samesex + black + hispan, data=babyData)
lh1 = linearHypothesis(flm1,”samesex = 0”, test=”F”)

# First Stage Test in AE w/ Multiple Instruments
flm2 = feols(morekids ~ boyBoy + girlGirl + black + hispan, data=babyData)
lh2 = linearHypothesis(flm2,c(”BoyBoy”,”GirlGirl”), test=”F”)
```

Results:

- With “Same Sex”: $F = 25.874$
- With “BoyBoy” and “GirlGirl”: $F = 12.94$

Checking The IV Assumptions II

- The two major assumptions:
  1. Relevance: $Cov(X, Z) \neq 0$, or that $\mathbb{E}\left[\mathbf{z}_i \mathbf{x}_i'\right]$ has full rank
  2. Exogeneity: All instruments must be uncorrelated with ε
- Now, let's talk about the exogeneity assumption.
- With one IV, it's impossible to test exogeneity, hence the need to evaluate exogeneity by thinking about whether some unobserved variables might be correlated with the instrument.
- Two popular tests:
  1. Hausman-Wu Test: Are $\widehat{\beta}_{IV}$ and $\widehat{\beta}_{OLS}$ the same?
  2. Sargan J Test: If we leave out z_i^* from our regression does $\mathbb{E}[(Y_i - X_i \widehat{\beta}_{2SLS})\, z_i^*] = 0$?
- These have fallen out of favor because they don't make sense if we have heterogeneous treatment effects (ie: β_i).
- Why? different instruments identify different complier groups.

Overidentification

- When the number of regressors equals the number of instruments, the model is **just identified**. When there are more instruments than regressors, the model is **overidentified**.
- For a just identified model, the 2SLS estimator simplifies to

$$\mathbf{b}_{IV} = \left(\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{Z}'\mathbf{y},$$

which is often called simply the **IV estimator**.

A Note on Moments

- Recall the property of the OLS estimator:

$$\mathbf{X}'\mathbf{e}_{OLS} = \mathbf{0}, \text{ with } \mathbf{e}_{OLS} = (\mathbf{y} - \mathbf{X}\mathbf{b}_{OLS})$$

- Similarly, for the IV estimator,

$$\mathbf{X}'\mathbf{e}_{IV} = \mathbf{0}, \text{ with } \mathbf{e}_{IV} = (\mathbf{y} - \mathbf{X}\mathbf{b}_{IV})$$

- A consequence of this is that we can't test the assumption that the instrument(s) and error terms are uncorrelated by looking at the correlation between the instrument(s) and residuals.
- Similarly, the residuals from OLS don't allow us to test whether the error terms are correlated with the regressors.
- Stories about an IV being excluded are just a story.

Tests of Exogeneity?

One possibility is the Durbin-Wu-Hausman Test which compares $\Delta = \hat{\beta}_{OLS} - \hat{\beta}_{2SLS}$ under the null that X is exogenous (Z is assumed exogenous).

$$\begin{aligned} H_0 &: \beta_{OLS}, \beta_{2SLS} \text{ are consistent}, \quad \beta_{OLS} \text{ is efficient} \\ H_1 &: \beta_{2SLS} \text{ is consistent}, \beta_{OLS} \text{ is not} . \end{aligned}$$

This gives a chi-squared test statistic:

$$T = \Delta^T \left(V(\hat{\beta}_{2SLS}) - V(\hat{\beta}_{OLS}) \right)^{\dagger} \Delta \sim \chi^2_K$$

This uses the pseudo inverse of the variance matrix where K is the rank of that matrix.

Idea: Reject H_0 if $\|\Delta\| \gg 0$ (2SLS and OLS disagree).

The other possibility is the Sargan J-Test which exploits **overidentification** (more instruments that endogenous regressors)

1. Pick an instrument(s) $z_i^* \in \mathbb{R}^{N \times L}$ to drop.
2. Run 2SLS using remainder except z_i^* to get $\widehat{\beta}^*_{2SLS}$ with K restrictions.
3. Evaluate the moment condition: $g_n(\widehat{\beta}^*_{2SLS}) = \mathbb{E}[(Y_i - X_i \widehat{\beta}^*_{2SLS})\, z_i^*]$.
4. Construct the quadratic form: $J = g_n(\widehat{\beta}^*_{2SLS})^T \, W \, g_n(\widehat{\beta}^*_{2SLS}) \sim \mathrm{X}^2_{L-K}$ where $L - K$ are the number of extra moments (dimension of z_i^*).

Software reports all three

```
> feols(worked ~ morekids + black + hisp +othrace + age + agefst |
+           samesex ~ boy1st + boy2nd, data=babydata)
TSLS estimation - Dep. Var.: worked
                  Endo.    : samesex
                  Instr.   : boy1st, boy2nd
Second stage: Dep. Var.: worked
Observations: 5,000
Standard-errors: IID
             Estimate Std. Error   t value   Pr(>|t|)
(Intercept)  0.645227   0.236454  2.728761 6.3795e-03 **
fit_samesex -0.300863   0.403954 -0.744795 4.5643e-01
morekids    -0.128419   0.037575 -3.417631 6.3674e-04 ***
black        0.194002   0.032675  5.937304 3.0934e-09 ***
hisp         0.056280   0.045018  1.250171 2.1130e-01
othrace      0.072025   0.052196  1.379900 1.6768e-01
age          0.019904   0.003423  5.814619 6.4551e-09 ***
agefst      -0.025480   0.003559 -7.158666 9.3305e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.509524   Adj. R2: 0.043844
F-test (1st stage), samesex: stat = 3.21083, p = 0.040406, on 2 and 4,991 DoF.
                Wu-Hausman: stat = 0.58365, p = 0.444923, on 1 and 4,991 DoF.
                    Sargan: stat = 0.32724, p = 0.567286, on 1 DoF.
```

1. We really want strong instruments that are good predictors of the endogenous X's.
2. Everyone expects us to show the first-stage F-statistic telling us whether excluded instruments predict endogenous X beyond the exogenous regressors.
    - With appropriate clustering/standard errors we want to see $F > 10$. This is large enough in practice but probably not in theory!
    - Ideally we'd have a much stronger F-stat but good IV are hard to come by
3. We can't really test exclusion or exogeneity of your instruments.
    - This is largely a story that comes from some economic theory or understanding of the world.
    - There are tests but they don't show what we really care about
        - Wu-Hausman shows us if $\beta_{IV} = \beta_{OLS}$ (not if instruments are valid)
        - Sargan shows us if different instruments give the same $\widehat{\beta}_{2SLS}$ (which might be because of heterogeneous treatment effects).

Weak instruments after 2020: $F > 10$ was never enough

- Staiger–Stock's $F > 10$ keeps the *bias* of 2SLS under 10% of the OLS bias. It says nothing about the size of the t-test.
- Lee, McCrary, Moreira and Porter (2022): with one instrument, the 5% 2SLS t-test can reject a true null up to 15% of the time at $F \approx 10$. Guaranteeing 5% size with the usual critical value needs $F > 104.7$; their tF procedure inflates the critical value with the observed F instead.
- Simulated, one instrument, $\rho = 0.95$, true $\beta = 0$:

Mean first-stage F	2SLS t-test, actual size	Anderson–Rubin test, actual size
11	0.094	0.045
105	0.049	0.052

The t-test over-rejects at $F \approx 10$ and is fine at $F \approx 100$; Anderson–Rubin is fine at both.

Anderson–Rubin: inference that does not care how weak the instrument is

- To test $H_0 : \beta = \beta_0$, form $y_i - \beta_0 x_i$ and regress it on the instrument. Under H_0 and exclusion, the coefficient is zero *whatever the first stage looks like*: the test never divides by the first stage, so weakness cannot distort it.
- Invert it: the AR confidence set is the set of β_0 not rejected. With a strong instrument it is close to the usual interval; with a weak one it is wide, possibly unbounded, and that is the honest answer. (A 95% interval that is always finite cannot be valid under weak instruments — Dufour 1997.)
- Just-identified, one endogenous variable: this is a grid search over β_0 with one regression each, a dozen lines of code (workbook).
- Practical rule for this course: report the first-stage F; if it is below ≈ 100, also report the AR interval. `ivDiag` and `ivmodel` in R do it; `linearmodels` in Python.

Local average treatment effects

When effects differ across people, what does IV estimate?

- Session 6 wrote $y_i = \beta_0 + \beta_1 D_i + \varepsilon_i$ with *one* β_1. In business data the effect of a loan, an app, a refund, a training program differs across customers, and the ones who take it up are not a random draw.
- Setup (Imbens and Angrist 1994): binary treatment D_i, binary instrument Z_i. Two sets of potential outcomes: potential treatments $D_i(1), D_i(0)$ (what i does if $Z_i = 1$ or 0) and potential outcomes $Y_i(1), Y_i(0)$.
- Running example: a retailer emails a random half of its customers (Z_i) encouraging them to install the loyalty app (D_i). Outcome: spending next quarter. Some customers would install it anyway; some never will; some install it *because* of the email.
- Readings: Hansen Ch. 12.29–12.31; Goff Ch. 4; Mixtape Ch. 7.

Four types of customer, two of which you cannot see

	$D_i(0) = 0$	$D_i(0) = 1$
$D_i(1) = 0$	never-taker	defier
$D_i(1) = 1$	complier	always-taker

- Always-takers install the app with or without the email. Never-takers do not install it either way. Compliers install it only if emailed. Defiers install it only if *not* emailed.
- We observe (Z_i, D_i), never the pair $(D_i(0), D_i(1))$. A customer with $Z_i = 1, D_i = 1$ is an always-taker or a complier; we cannot tell which. That is the whole difficulty.
- Three assumptions:
  1. Independence: $Z_i \perp (Y_i(1), Y_i(0), D_i(1), D_i(0))$. The email is randomized.
  2. Exclusion: Z_i affects Y_i only through D_i. The email itself does not change spending.
  3. Monotonicity: $D_i(1) \geq D_i(0)$ for everyone. No defiers.

The LATE theorem in three lines

Write $D_i = D_i(0) + (D_i(1) - D_i(0))Z_i$ and $Y_i = Y_i(0) + (Y_i(1) - Y_i(0))D_i$.

$$\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0] = \mathbb{E}\big[(Y_i(1) - Y_i(0))\,(D_i(1) - D_i(0))\big] \quad \text{(independence}$$

$$= \mathbb{E}\big[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)\big] \cdot \mathbb{P}\big(D_i(1) > D_i(0)\big) \quad \text{(monotonicity:}$$

$$\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0] = \mathbb{P}\big(D_i(1) > D_i(0)\big) \quad \text{(the first stage}$$

Divide:

$$\beta_{IV} = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]} = \mathbb{E}\big[Y_i(1) - Y_i(0) \mid \text{complier}\big] \equiv \text{LATE}.$$

- The Wald estimator from Assignment 3, now with a population it belongs to: the compliers.
- Always-takers and never-takers drop out of the numerator because $D_i(1) - D_i(0) = 0$ for them: the email did not change what they did, so it cannot reveal what the app did for them.
- Without monotonicity, defiers enter with a minus sign and the ratio is a difference of two effects.

$10{,}000$ customers: 20% always-takers (effect of the app 2.0, and they spend more anyway), 40% compliers (effect 0.8), 40% never-takers (effect 0.5 if they ever installed it). Email to a random half.

Quantity	Estimate	What it is
OLS, y on D	2.72 (0.03)	effect on the treated $+$ selection (always-takers spend more anyway)
Reduced form, y on Z	0.32 (0.04)	intent-to-treat: effect of the *email*
First stage, D on Z	0.39 (0.01)	share of compliers (truth 0.40)
2SLS $=$ Wald ratio	0.80 (0.09)	LATE: effect for compliers (truth 0.80)
ATE (truth)	0.91	average over everyone, including never-takers
ATT (truth)	1.40	average over app users: always-takers and encouraged compliers
Complier mean tenure	1.99	Abadie's κ estimate (truth 2.01; always-takers 4.00, never-takers 1.00)

Reading that table

- OLS compares app users to non-users; always-takers are heavy spenders regardless, so it is off by a factor of three.
- Reduced form is the effect of the *email*: what the marketing team cares about, and it needs no exclusion restriction.
- First stage is the complier share: 40% of customers can be moved by an email.
- 2SLS recovers the compliers' effect, 0.8: not the ATE (0.91), not the ATT (1.40). No estimator from this experiment delivers those, because it never moved the other two types.
- Who are the compliers? Abadie (2003): for any covariate X, $\mathbb{E}[X \mid \text{complier}] = (\mathbb{E}[XD \mid Z = 1] - \mathbb{E}[XD \mid Z = 0])/(\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0])$, the theorem applied to XD. The last row recovers the compliers' mean tenure. Describe your compliers.

Different instruments, different compliers, different LATEs

- A second valid instrument (a $5 credit for installing) moves a different set of customers: price-sensitive ones rather than email-responsive ones. Its LATE is the effect for *those* compliers.

- Two valid instruments can give two different 2SLS estimates, and the overidentification test from earlier today can reject with both instruments perfectly valid. The J test assumes one β.

- Which LATE do you want? The one whose compliers match the policy. “Should we send more emails?” wants the email LATE. “Should we make the app opt-out?” wants an effect on today’s never-takers, which no encouragement design identifies.

- The LATE framework is Vytlacil (2002)’s point in reverse: monotonicity is exactly the assumption that treatment is chosen by a single index crossing a threshold, $D_i = \mathbb{1}\{v_i \leq \pi(Z_i)\}$, which is how every selection model in this course has been written.

When monotonicity is the assumption to worry about

- Encouragement that backfires. A reminder that annoys some customers into uninstalling: defiers.

- Angrist and Evans (1998). Two same-sex children as an instrument for a third child: parents who want two boys become defiers when they get two girls.
- Judges and examiners, next section: a judge who is strict on drug cases and lenient on property cases is lenient for some defendants and strict for others. Monotonicity across judges is the live question in that literature.
- De Chaisemartin (2017): with “a few” defiers, 2SLS still estimates a LATE for a subpopulation of compliers, provided compliers outnumber defiers with the same effects. The theorem is tolerant, not fragile.
- Exclusion failing is the more common problem in practice, and it is not testable. Ask what else the instrument could do.

Judge and examiner designs

Random assignment to a decision-maker

- Many treatments are *decided by a person*: a judge sets bail, a loan officer approves credit, a claims adjuster pays or denies, a moderator removes a post, a recruiter advances a candidate, a customer-service agent grants a refund.
- Decision-makers differ in strictness. If cases are assigned to them at random, then which decision-maker you drew is a random shock to your treatment probability that has nothing to do with you.
- Instrument: the assigned decision-maker's leniency, measured as their decision rate on *other* cases (leave-one-out).
- Canonical: Kling (2006, incarceration), Dobbie and Song (2015, bankruptcy), Maestas, Mullen and Strand (2013, disability insurance), Dobbie, Goldin and Yang (2018, pretrial detention); in business, Dobbie et al. (2021, loan officers). Assignment 4 has loan officers; today, refund requests.

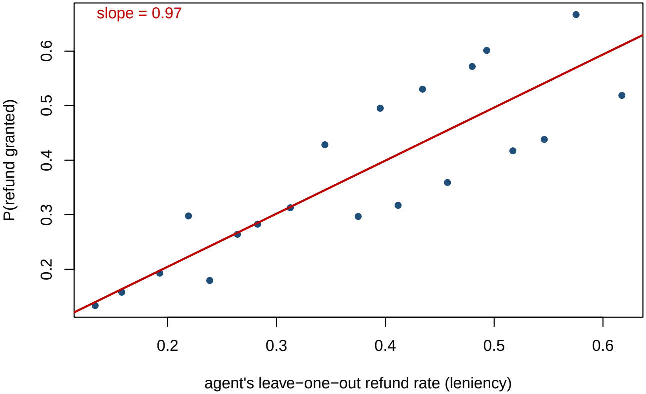
Running example: refund requests

- $20{,}000$ refund requests routed at random to 50 customer-service agents. Agent j grants a refund if the claim's merit q_i (which the agent sees and we do not) exceeds the agent's personal threshold. Agents differ: some grant 20% of requests, some 60%.
- Outcome: the customer's spending over the next year. Truth:

$$\text{spend}_i = 100 + \tau_i R_i + 30\, q_i + \varepsilon_i, \qquad \tau_i = 20 - 15\, q_i.$$

  A refund builds the most goodwill for customers with *weak* claims (they expected to lose); customers with strong claims also spend more regardless, since good customers make good claims.
- Question for the business: what does granting a refund do to future spending? Naive answer: compare spending of refunded and non-refunded customers.

The first stage is the whole design



Binned: the probability a request is granted against the assigned agent's leave-one-out grant rate. A slope near one says a request routed to an agent who grants 10 points more often is 10 points more likely to be granted. This picture is the first thing every judge-design paper shows.

	OLS, spend on refund	First stage, refund on leniency	2
Coefficient	47.9	0.97	1
	(0.6)	(0.004)	(
Truth	ATE $= 19.9$; effect on refunded $= 7.2$	—	LATE (margina
First-stage F		1773	
First stage by claim size	small 0.86, medium 0.95, large 1.10 (all positive: no sign of defiers)		

- OLS is off by a factor of three: refunded customers had better claims and spend more anyway.
- 2SLS gives the effect for marginal claims: those whose outcome depends on which agent they drew. It sits below the ATE because marginal claims have moderate merit and the goodwill effect is largest for weak ones. The effect on customers who actually got refunds is smaller still.

The three conditions, in this setting

- Relevance. Agents differ and the leave-one-out rate predicts the decision (first stage, F). Leave-one-out keeps the case's own decision, and so its own q_i, out of its instrument.
- Exclusion. The agent affects spending *only* through the decision. Fails if lenient agents are also friendlier, faster, or offer store credit. Random routing buys independence, not exclusion.
- Monotonicity. Any request a strict agent grants, a lenient one would too. Fails if agents rank claims differently (generous on shipping, strict on damage, or the reverse): then "lenient" is not one dimension.
- Testing it (Frandsen, Lefgren and Leslie 2023): first stage positive in every subgroup (claim type, size, channel); reduced form monotone in leniency. Our table passes for small, medium, and large claims. Necessary, not sufficient.
- With many agents, 2SLS on continuous leniency is a weighted average of pairwise LATEs (Imbens and Angrist 1994).

Practical details that decide whether the design works

- Is assignment really random? Rotations, caseload balancing, and specialization break it. Regress case characteristics on leniency: nothing should predict it (the experiment's balance test).
- Leniency is estimated. Few cases per agent makes it noisy and pulls 2SLS toward OLS (many weak instruments). Leave-one-out or split-sample leniency; agents with enough cases.
- Cluster by decision-maker. Cases sharing an agent share the instrument.
- Whose effect? Marginal cases: right for a policy that moves everyone's threshold ("grant all refunds under $50"), wrong for one that changes what always-granted cases receive.
- Business uses. Any process with random routing and human discretion is an experiment you did not have to run: refunds, credit lines, moderation, fraud holds, rep-offered discounts.

Shift-share designs

Exposure designs

- A national (or global) shock hits local units in proportion to their pre-existing exposure:

$$B_m = \sum_k s_{mk}\, g_k, \qquad s_{mk} = \text{unit } m\text{'s share in category } k \text{ (predetermined)}, \quad g_k = \text{shock to categ}$$

- Bartik (1991): local employment growth from national industry growth and local industry mix. Autor, Dorn and Hanson (2013): local exposure to Chinese imports. Now everywhere: banks' sector mix $\times$ sector credit shocks; retailers' category mix $\times$ national category demand; firms' country mix $\times$ tariffs.
- Use B_m as an instrument for the local variable it predicts, and estimate that variable's effect on a local outcome.
- Running example: markets differ in which sectors their banks lend to; national sector credit shocks g_k move local credit growth x_m; the question is the effect of local credit on firm entry y_m. A local demand shock η_m moves both. Truth: 0.4.

Why would B_m be uncorrelated with η_m? Two different arguments, and they need different evidence.

- Shares are exogenous (Goldsmith-Pinkham, Sorkin and Swift 2020). The instrument is really the vector of shares; the shocks weight them. Valid if a market's sector mix is unrelated to its local shocks, conditional on controls: a differences-in-differences claim. Evidence: pre-trends and balance on the shares that matter (the Rotemberg weights say which).
- Shocks are exogenous (Borusyak, Hull and Jaravel 2022). Shares can be anything; the national shocks g_k are as good as random across many categories. Evidence: many independent shocks, balance at the shock level.
- Same instrument, same number, different justifications and failure modes. Say which one you rely on. Assignment 4 breaks the first and shows the second surviving.

Which sectors carry the estimate: Rotemberg weights

Sector	shock g_k	Rotemberg weight α_k	just-identified $\hat{\beta}_k$
2	-1.42	0.39	0.48
4	1.56	0.25	0.40
6	1.24	0.15	0.17
1	1.12	0.12	0.66
5	-0.56	0.07	0.47
3	0.18	0.00	4.39
2SLS with B_m		$\sum_k \alpha_k = 1$	0.44 $= \sum_k \alpha_k \hat{\beta}_k$

Goldsmith-Pinkham, Sorkin and Swift: the shift-share 2SLS estimate is a weighted average of the just-identified estimates you would get using each sector's exposure $s_{mk} g_k$ alone, with weights $\alpha_k \propto g_k \sum_m s_{mk} \tilde{x}_m$. Weights can be negative. Here one sector carries most of the estimate; that is the sector whose share-exogeneity claim needs defending, and its own estimate is what to compare against the pooled one.

- $1{,}000$ markets, 6 sector shocks: the instrument varies through six numbers. Markets sharing a dominant sector have nearly the same B_m and, with any sector-level component in the outcome, correlated errors.
- Adão, Kolesár and Morales (2019): market-level robust SEs treat $1{,}000$ instrument values as independent and can be wildly overconfident.

Design	Nominal size	Actual rejection rate under H_0
Market-level robust SE, sector shocks in the outcome	0.05	0.73

- A nominal 5% test rejects a true null most of the time. Fix: standard errors at the *shock* level (AKM, or the Borusyak–Hull–Jaravel shock-level regression), which with six shocks means honest, wide intervals. The effective sample size is the number of shocks, not units.

In practice

```r
# LATE: an encouragement design
feols(y ~ 1 | D ~ Z, d)                                 # 2SLS = Wald ratio
feols(y ~ Z, d); feols(D ~ Z, d)                       # reduced form, first stage (complier share)
(mean(x[Z==1] * D[Z==1]) - mean(x[Z==0] * D[Z==0])) / (mean(D[Z==1]) - mean(D[Z==0]))   # E[x | complier]
# Judges: leave-one-out leniency, cluster by agent
tot <- tapply(R, agent, sum); n_j <- tapply(R, agent, length); len <- (tot[agent] - R) / (n_j[agent] - 1)
feols(spend ~ 1 | R ~ len, d, cluster = ~agent)
# Shift-share: build B and instrument
B <- as.vector(shares %*% g); feols(y ~ 1 | x ~ B, d, vcov = "hetero")
```

- Shock-level standard errors: `ssaggregate` (Borusyak–Hull–Jaravel) or the AKM code; Rotemberg weights: `bartik_weight` (GPSS). Both are Stata-first; the R ports are on GitHub. The workbook computes Rotemberg weights by hand in five lines.

Recap

- With heterogeneous effects, IV estimates the LATE: the effect for compliers, the people the instrument actually moved. Reduced form over first stage; first stage is the complier share; describe the compliers with Abadie's trick.
- Judge designs: random routing to decision-makers of different strictness. Leave-one-out leniency, first-stage picture, monotonicity across decision-makers is the assumption to test, cluster by decision-maker. Any human-discretion process with random assignment qualifies.
- Shift-share: exposure to national shocks. Decide whether you believe the shares or the shocks; Rotemberg weights say which sectors matter; standard errors belong at the shock level.
- Weak instruments: $F > 10$ was never enough; report Anderson–Rubin.

`inclass_session7.Rmd`:

1. Reproduce the encouragement design; change the complier effect and watch which rows of the table move.
2. Reproduce the refund design; give agents type-specific generosity (lenient on shipping, strict on damage) and re-run the subgroup first stages.
3. Reproduce the shift-share estimate and Rotemberg weights; then drop the top-weight sector and see what happens to the estimate and the first-stage F.